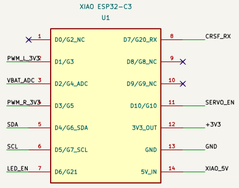
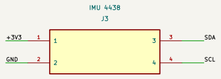
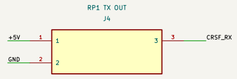
BUOM C3 / REV A / 2S HV SERVO CARRIER

Prototype: electrical CAD verified; flight and thermal performance unmeasured.















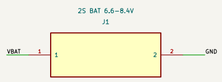


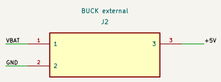


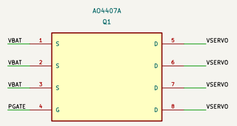






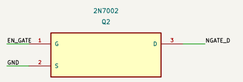














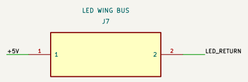


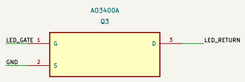










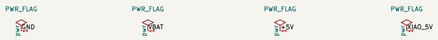












U1 BAT pads NC. NEVER connect 2S to XIAO BAT or 5V.
J2: external regulated 5V >= 1A; common GND.
D1 prevents USB feeding the external 5V bus.
J4: receiver TX -> MCU RX. Receiver RX unconnected.

J7: 40 parallel LED branches, EACH with its own 2k2 resistor.
White LED: +5V -> 2k2 -> anode; cathode -> LED_RETURN.
IMU at CG: X forward, Y left, Z up; leads <= 50mm.
Digital signal levels are 3.3V except buffered servo PWM.

Q1 source pins 1/2/3 -> VBAT; drains 5/6/7/8 -> VSERVO.
Gate pulled to source by R3: servo power OFF at reset.
PWM is forced LOW before switching servo rail OFF.
Use keyed battery cable; no reverse-polarity protection.

Do not fit a 1S-only servo or SG90/MG90S.
Bench checks: polarity, 5V regulation, 3V3, ADC scale,
servo pulse range, idle-off rail, current peaks, RF failsafe.

Sheet: /
File: buom_c3.kicad_sch

Title: Buom C3 carrier

Size: A2 | Date: 2026-11-12 | **Rev: A – prototype**

KiCad E.D.A. 10.0.8 | Id: 1/1